AIR FORUM 2026 · POSTER 19 · MAY 27, 12:30–1:30 PM · WASHINGTON D.C.

ST JOHN'S College

Starting *Small* with AI: Building Trustworthy KPI Dashboards

Elmer Yglesias · Chief Data Officer, St. John's College · Elmer.Yglesias@sjc.edu · elmerdata.ai

Not a BI tool. *A governance-first AI validation layer for institutional KPI pipelines — free, open-source, and built for lean IR offices.*

THE 10-STEP VALIDATION FRAMEWORK

I · DEFINE — *Describe the KPI precisely*

01	**Register Metadata**	All required fields present and non-empty
02	**Validate Definition**	Formula operator · Time period · Definition precision
03	**Entity Coherence**	Cross-system join risk: Low / Moderate / High

II · VALIDATE — *Test for quality and governance*

04	**Data Quality**	Outliers ($z \geq 2.5\sigma$) · Step changes (≥25%) · Flags only, never auto-corrects
05	**AI Use Assessment**	Documents permitted techniques · Excludes black-box scoring
06	**Governance Check**	Owner · Steward · Limitations · Equity risks · Signoff HARD GATE
07	**Readiness Decision**	Ready for Dashboard Use · or · Not Ready PUBLISH GATE
08	**Summary Report**	JSON audit trail · CSV executive summary · PNG time-series chart

III · ENHANCE — *Advisory: design quality and standards*

09	**Design Quality**	SMART (5 criteria) + Tufte visualization principles (5 criteria) ADVISORY
10	**Industry Standards**	EDUCAUSE · AGB · NACUBO alignment · Score: X/9 ADVISORY

"Dashboard quality cannot exceed KPI governance quality. The 10-step framework is the engine. AI is the power steering."

THREE-MODULE PIPELINE

kpi_parser.py
Claude API
Plain-language intake · Human review before validation
→ parsed_kpi.json

kpi_readiness_10step.py
Pure Python
10-step engine · No API key required
→ _report.json
→ _summary.csv
→ _chart.png

kpi_improver.py
Claude API
Refinements · HTML dashboard · LLM survey analysis
→ _dashboard.html
→ _suggestions.txt

STEP 9 — DUAL ASSESSMENT FRAMEWORK

SMART Doran, 1981	Metric construction — is the KPI specific, measurable, achievable, relevant, and time-bound?
Tufte / Few 2001 · 2006	Metric interpretation — will it chart clearly, honestly, and with sufficient comparison context?
Governance Steps 1–6	Institutional accountability — ownership documented, human signoff required before publication.

GOVERNANCE AT EVERY TOUCHPOINT

- ✓ Human review before validation runs
- ✓ Incomplete governance blocks publication
- ✓ Equity risks required in every report
- ✓ Anomalies flagged — never auto-corrected
- ✓ All AI output labeled as draft for review
- ✓ LLM-agnostic — swap any API provider

NORCO COLLEGE PILOT — 13 KPIS REVIEWED

- 10/13 KPIs lacked a comparison benchmark or target
- 2 KPIs showed significant trend changes with no annotation or explanation
- All 13 were displayed on separate pages, obscuring cross-metric relationships

Conducted with Hayley Ashby, Ed.D., Norco College (RCCD).
Presented at REACH Research Analytics Summit, Newport RI, April 2026.

Scan for code, README & supplementary materials

github.com/ElmYgl/kpi_readiness_10step

Presented at AIR Forum 2026, Washington D.C.
Proposal #2234062
Elmer.Yglesias@sjc.edu · elmerdata.ai
ORCID: 0009-0004-9538-2159

Get Started in Minutes

Three entry levels · Any IR office · No vendor dependency required for core validation

github.com/ElmYgl/kpi_readiness_10step
Full paper: AIR Forum 2026 · Proposal #2234062
Elmer.Yglesias@sjc.edu · elmerdata.ai

INSTALLATION & SETUP

Level 1 Core Validation Only

Requires: Python 3.8+ only · No API key · No additional install

```
# Download from GitHub, then:
python kpi_readiness_10step.py
```

Runs demo immediately. Generates _report.json and _summary.csv. Edit the KPIInput block in main() to validate your own KPI.

Level 2 With Plain-Language Parser

Requires: Python 3.8+ · Claude API key · pip install anthropic

```
pip install anthropic
# Windows PowerShell — set API key:
[System.Environment]::SetEnvironmentVariable(
  'ANTHROPIC_API_KEY', 'sk-...', 'User')
# Mac / Linux:
export ANTHROPIC_API_KEY='sk-...'
python kpi_parser.py
```

Paste any KPI description in plain language. Three data options: include in description, upload Excel/CSV, or no data yet. Human review required before validation runs.

Level 3 Full Pipeline

Requires: Python 3.8+ · Claude API key · All three modules

```
pip install anthropic openpyxl matplotlib
python kpi_parser.py
# After validation completes:
python kpi_improver.py
```

kpi_improver.py auto-finds the latest _report.json and generates the board-ready HTML dashboard and AI-suggested refinements.

EVERY PIPELINE RUN PRODUCES

_report.json	Full 10-step audit trail
_summary.csv	Executive summary · Opens in Excel
_chart.png	Time-series with anomalies marked
_dashboard.html	Board-ready · Opens in any browser
_suggestions.txt	AI refinement candidates for review

White = kpi_readiness_10step.py (pure Python) · Gold = kpi_improver.py (Claude API)

AI GOVERNANCE — STEP 05

✓ PERMITTED	✗ EXCLUDED
✓ Entity resolution across systems	✗ Automated individual decisions
✓ Anomaly detection on aggregates	✗ Black-box scoring
✓ NLP for survey comments	✗ Auto-correction without review
✓ Synthetic data — clearly labeled	✗ Ungoverned AI output published

SESSION LEARNING OUTCOMES

LO1 Identify which AI techniques are appropriate for KPI validation — and which must be excluded — with documented rationale

LO2 Apply the three-entry-level pipeline to begin validating KPIs within your own office context

LO3 Implement a reusable 10-step validation framework aligned with EDUCAUSE, AGB, and NACUBO standards

DEMO RUN — FIRST-YEAR RETENTION RATE

Step 07 **READY ✓** Dashboard Use

Step 09.1 5/5 SMART — all criteria met

Step 09.2 4/5 Tufte — add benchmark

Step 10 8/9 Strong Alignment

Tufte (Step 9.2) and AGB (Step 10) flag the same gap: missing comparison benchmark. Convergent finding — two independent frameworks, one recommendation.

CITE THIS WORK

Yglesias, E. (2026). Starting small with AI: Building trustworthy KPI dashboards. Paper presented at *AIR Forum 2026*, Washington, D.C.
github.com/ElmYgl/kpi_readiness_10step

EDUCAUSE Data Governance Framework 2023

AGB Dashboard Metrics for Boards 2022

NACUBO KPI Framework 2024

KEY REFERENCES

Ashby, H., & Yglesias, E. (2026, April). From strategy to screens: Designing dashboards that support decision-making [Presentation]. REACH Research Analytics Summit, Newport, RI.
Doran, G. T. (1981). There's a S.M.A.R.T. way to write management's goals and objectives. *Management Review, 70*(11), 35–36.
EDUCAUSE. (2023). *Data governance framework.* EDUCAUSE.
EDUCAUSE. (2024). *2024 EDUCAUSE AI landscape study.* EDUCAUSE.
Few, S. (2006). *Information dashboard design.* O'Reilly.
McGuire, L. (2017). Institutional data quality and the data integrity team. *AIR Professional Files, Summer 2017.* Association for Institutional Research.
NACUBO. (2024). *KPI framework for higher education.* NACUBO.
NIST. (2023). *Artificial intelligence risk management framework (AI RMF 1.0).* NIST AI 100-1. U.S. Department of Commerce.
Teshome, S. (2025). Key performance indicators in higher education: A systematic review. *International Journal of Education, Management, and Technology.*
Tufte, E. R. (2001). *The visual display of quantitative information* (2nd ed.). Graphics Press.
Association of Governing Boards. (2022). *Dashboard metrics for boards.* AGB.
Webber, K. (2023). AI in higher education: Implications for institutional research. *eAIR Newsletter.* Association for Institutional Research.
Yglesias, E. (2026). Starting small with AI: Building trustworthy KPI dashboards. Paper presented at *AIR Forum 2026*, Washington, D.C.

Elmer Yglesias · St. John's College · Elmer.Yglesias@sjc.edu
elmerdata.ai · ORCID: 0009-0004-9538-2159